Dataset & Fusion

RGBT-Tiny Object Detection: Final Evaluation Report

Github Repo:

https://github.com/BillMentzinis/CV_Final_Project

1. Dataset & Methodology

The project utilizes the RGBT-Tiny dataset, a large-scale, multi-modal benchmark engineered for the detection of extremely small objects. It consists of over 93,000 paired Visible and Thermal frames with 1.2 million manual annotations. The dataset is uniquely challenging as over 81% of targets are smaller than 16 x 16 pixels.

- Evaluation Parameters:
- Metric: mAP@0.5 and mAP@0.5:0.95 (COCO standard).
- Confidence Threshold: 0.001 (to ensure a full PR curve sweep for correct mAP computation).
- NMS IoU Threshold: 0.6 (Ultralytics default).
- Image Size: 640x640.
- Day/Night Split: 6,986 day images and 5,604 night images.
- Day/Night Classification: Defined by mean RGB brightness with a threshold of 90.5.

2. Evaluated Model Architectures

The following architectures were benchmarked to establish performance floors and test fusion efficacy:

- RGB Baseline: YOLOv8n trained on single-modality 3-channel RGB data.
- IR Baseline: YOLOv8n trained on single-modality 3-channel Infrared data.
- Early Fusion (Original): YOLOv8n modified for 4-channel input (RGB+IR concatenation).
- Early Fusion (Corrected): Incorporates three training fixes: channel-aware HSV augmentation (applied to RGB only), zero-initialization of IR first-conv weights, and extended patience (50 epochs).
- Mid Fusion (MLP): A dual-backbone YOLOv8n with an MLP fusion head processing 6-channel data.

3. Detailed Performance Results

mAP@0.5 Performance Summary

Model	Full Val	Day	Night	Day->Night Drop
RGB Baseline	0.406	0.497	0.236	-52.5%
IR Baseline	0.412	0.448	0.372	-16.9%
Early Fusion (Orig)	0.374	0.479	0.202	-57.8%
Early Fusion (Corr)	0.370	0.485	0.200	-58.8%
Mid Fusion (MLP)	0.390	0.481	0.239	-50.3%

Precision, Recall, and Localization (Full Val)

Model	Precision	Recall	mAP@0.5:0.95
RGB Baseline	0.573	0.399	0.171
IR Baseline	0.590	0.387	0.191
Early Fusion (Corr)	0.544	0.366	0.162
Mid Fusion (MLP)	0.559	0.385	0.158

4. Key Findings & Analysis

Finding 1: Validation of Sensor Hypotheses

The hypothesis that RGB performs better by day and IR better by night is confirmed. IR shows a resilient 16.9% drop at night, whereas RGB collapses by 52.5%. This is most dramatic for the ship class; ships have strong thermal contrast at night but are nearly invisible in RGB imagery.

Finding 2: The Fusion Paradox

Despite access to both sensors, fusion models failed to outperform the best single-modality model in any condition. At night, Early Fusion (~0.20) was worse than both baselines. The presence of degraded RGB channels actively hurts performance rather than being ignored. In night scenarios where an object is only visible in IR (like ships), RGB-heavy fusion approaches tend to collapse.

Finding 3: Structural Limitations of Fixed Weighting

Corrected training (zero-initialization and better augmentation) improved box localization quality but failed to raise the night performance ceiling of ~0.20 mAP@0.5. This confirms the failure is architectural—standard backbones learn fixed weights that 'average' across conditions, applying the same positive weight to noisy RGB pixels at night that it learned during the day.

5. Remaining Architectural Bottlenecks

- Conditional Weighting: Standard training cannot teach the model when to trust each modality.
- Model Capacity: YOLOv8n is nano-scale (3.2M parameters). Dual backbones have less specialization capacity.
- Class Imbalance: Classes like cyclist and pedestrian remain primary difficulties independent of modality.

Small Object Detection

Small Object Detection Using Yolov8

Our approach to improving small object detection follows a three-stage experimental pipeline. Each stage builds upon the previous one, progressively addressing the challenges of detecting tiny objects (as small as 4x4 pixels) in drone-captured aerial imagery. All three experiments share identical training hyperparameters to ensure a fair comparison, with the only differences being architectural modifications and inference strategies.

Training Hyperparameters (Shared Across All Experiments)
The following hyperparameters were used for both the Naive Baseline and P2 Baseline training. SAHI (Experiment 3) uses the trained P2 model without retraining.

General Training	
Parameter	**Value**
Epochs	500
Early stopping patience	20
Batch size	8
Image size	640
Initial learning rate	0.001
Momentum	0.937
Weight decay	0.0005
Warmup epochs	3
Warmup momentum	0.8
Warmup bias lr	0.1
Box loss	7.5
Classification loss	0.5
DFL loss	1.5
Dropout	0.1

Data Augmentation	
Mosaic	1.0
Mixup	0.1
HSV hue	0.015
HSV saturation	0.7
HSV value	0.4
Horizontal flip	0.5
Vertical flip	0.0
Scale	0.5
Shear	0.0
Erasing	0.4

Naive baseline

The first experiment establishes a performance baseline using the standard YOLOv8n architecture without any modifications. YOLOv8n is the nano variant of the YOLOv8 family, featuring 3.2M parameters and a three-scale detection head operating at strides 8, 16, and 32 (layers P3, P4, P5). For a 640-pixel input image, the smallest feature map produced is 80x64 at the P3 level (stride 8). This means that an object smaller than 8 pixels in the original image is represented by less than one cell on the finest detection grid, making it extremely difficult for the model to localize and classify.

The model was initialized from pretrained COCO weights (`yolov8n.pt`) and fine-tuned on the RGBT-Tiny RGB training split using the shared hyperparameters listed above. The purpose of this experiment is to quantify how well a standard, unmodified detector handles a dataset dominated by tiny objects, and to serve as a reference point for all subsequent improvements.

Detection heads	P3 (stride 8), P4 (stride 16), P5 (stride 32)
Finest feature map (640x512)	80x64

P2 baseline

The second experiment addresses the spatial resolution bottleneck identified in the naive baseline by adding a **P2 detection head at stride 4**. This architectural modification introduces a fourth detection scale that produces a 160x128 feature map for a 640x512 input, providing 4x more spatial resolution than the finest P3 layer. The key insight is that objects smaller than 8 pixels, which occupy less than one cell on the P3 grid, now span approximately 2-4 cells on the P2 grid, giving the network sufficient spatial information to localize and classify them.

The P2 head was integrated into the YOLOv8n architecture through a custom YAML configuration (`configs/model/yolov8n-p2.yaml`). The neck was extended with an additional upsampling path that concatenates backbone features from layer 2 (P2/4, 160x160) with the upsampled P3 features. The resulting P2 feature map is then processed through a C2f block and connected to the Detect module alongside the existing P3, P4, and P5 heads.

All training hyperparameters were kept identical to the naive baseline to isolate the effect of the architectural change. The model was built from the custom P2 configuration and initialized with pretrained backbone weights where applicable.

Architectural comparison		
Property	**Naive Baseline**	**P2 Mod**
Detection heads	P3, P4, P5	P2, P3, P4, P5
Finest stride	8	4
Finest feature map (640x512 input)	80x64	160x128
Spatial cells for 8px object	~1 cell	~2-4 cells

P2-mod and SAHI

The third experiment applies SAHI (Slicing Aided Hyper Inference) as an inference-time optimization on top of the trained P2 baseline model. Unlike the P2 head, which improves detection through architectural changes during training, SAHI operates exclusively at inference

time and requires no retraining. It addresses a complementary problem: even with a stride-4 detection head, the detector processes the entire 640x512 image at once, and very small objects may still lack sufficient pixel-level detail for confident detection.

SAHI works by:

1. Slicing the input image into overlapping patches of a fixed size.
2. Running the detector independently on each patch and on the full image.
3. Merging all predictions from all patches using a postprocessing strategy (e.g., NMS or GreedyNMM) to eliminate duplicate detections in overlapping regions.

We used the official SAHI library (‘*obss/sahi, https://github.com/obss/sahi’*). Since SAHI introduces several hyperparameters (slice size, overlap ratio, confidence threshold, postprocessing strategy), we conducted a systematic grid search to find the optimal configuration rather than relying on default values. The grid search was performed in two stages:

- **Stage 1** — **Slice size and overlap (16 combinations)**: Tested all combinations of slice sizes {128, 256, 320, 512} and overlap ratios {0.1, 0.2, 0.3, 0.4} at confidence threshold 0.001 (to match the baseline full PR-curve evaluation). All 16 combinations performed worse than standard inference.
- **Stage 1b — Confidence threshold discovery (6 combinations)**: Tested confidence thresholds {0.1, 0.25} with slice sizes {128, 512}. This revealed that confidence threshold is the most critical SAHI parameter. At conf=0.25 with slice=512, SAHI outperformed standard inference for the first time.
- **Stage 2 — Postprocessing fine-tuning (12 combinations)**: With slice=512, overlap=0.2, and conf=0.25 locked, tested all combinations of postprocess types {GREEDYNMM, NMS}, match metrics {IOU, IOS}, and match thresholds {0.3, 0.5, 0.7}.

Best SAHI configuration (found via grid search)	
Parameter	Value
Slice size	512x512
Overlap ratio	0.2
Confidence threshold	0.25
Postprocess type	GREEDYNMM

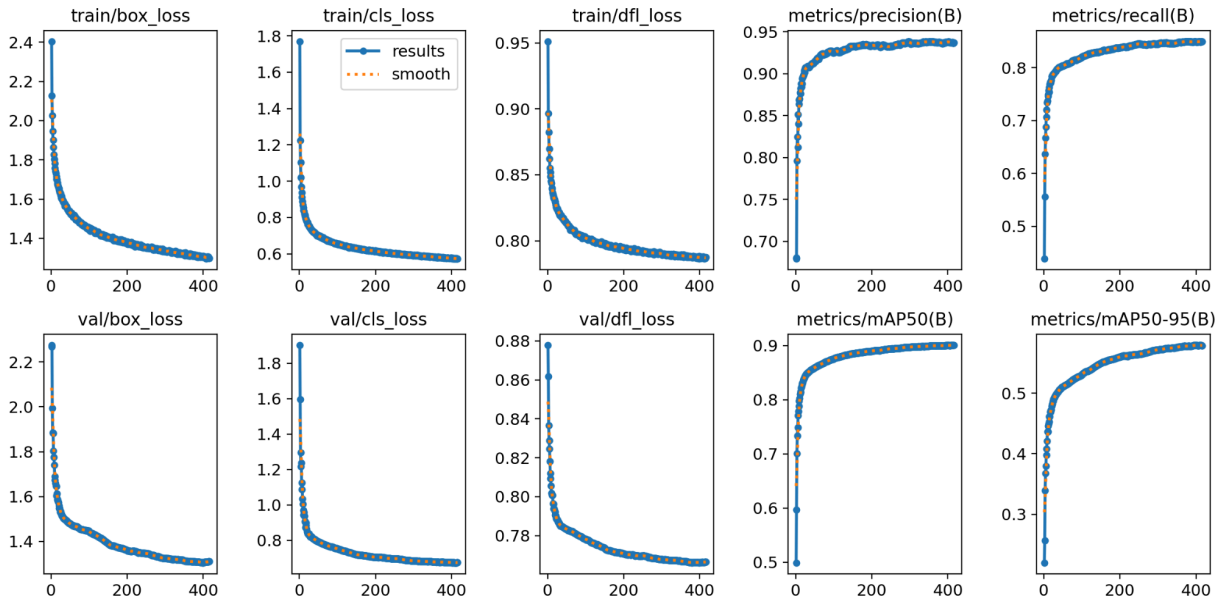
Best SAHI configuration (found via grid search)	
Postprocess match metric	IoU
Postprocess match threshold	0.7
Full-image prediction	Enabled

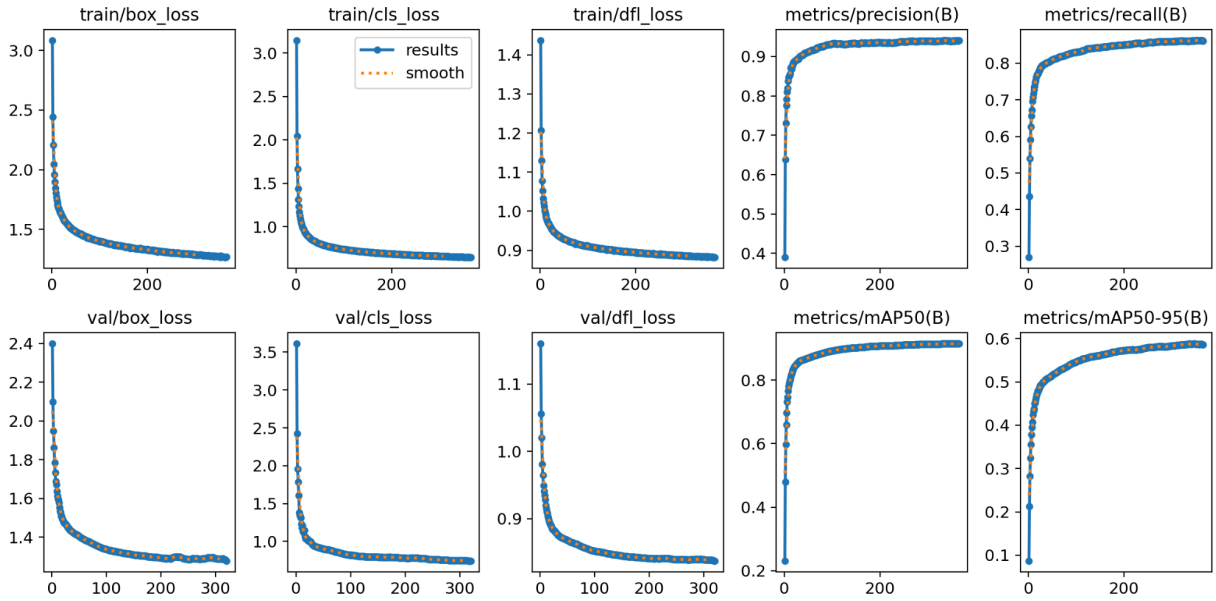
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Experiments & Results

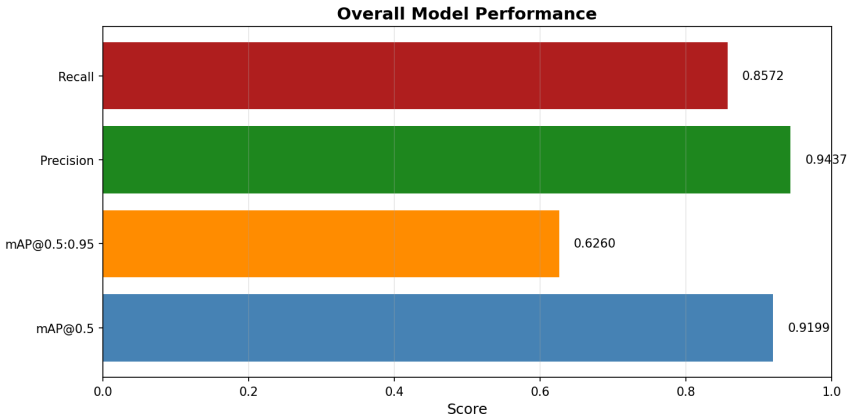
Naive Baseline vs P2 Baseline

Training metrics





Overall Metrics



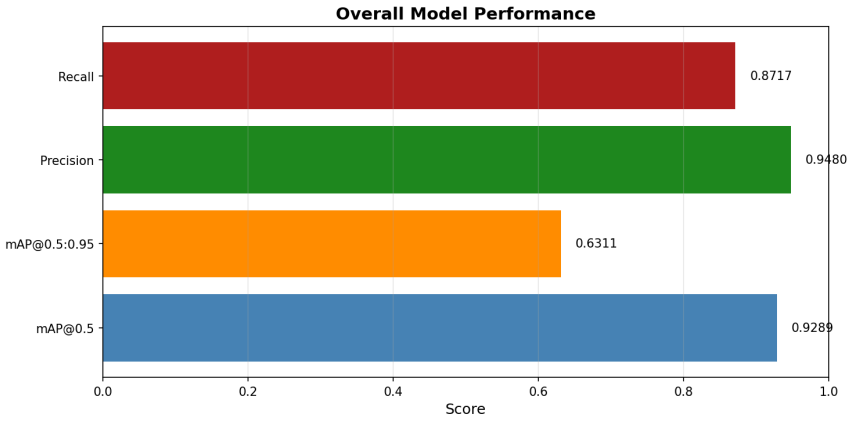
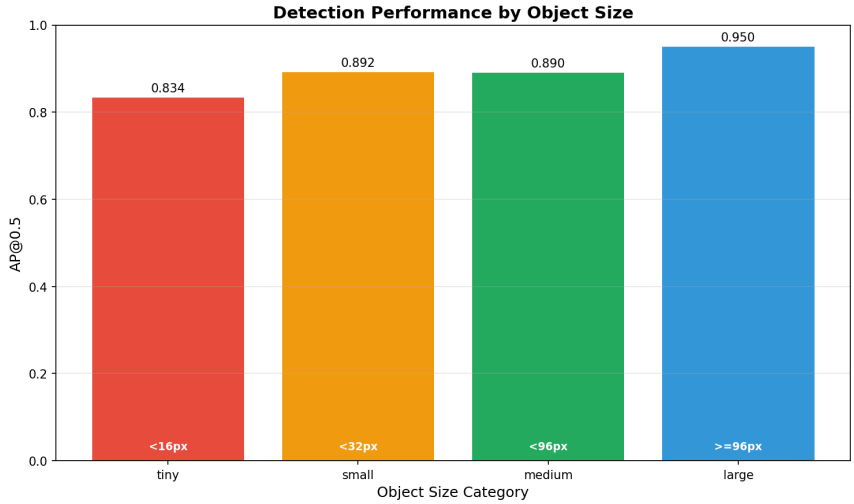


Figure Naive vs P2-mod

Overall metrics			
Metric	**Naive**	**P2-mod**	**Improvement**
mAp@0.5	0.9199	0.9289	+0.0090
mAP@0.5:0.95	0.6260	0.6311	+0.0051
Precision	0.9437	0.9480	+0.0043
Recall	0.8572	0.8717	+0.0145
Weighted mAP@0.5	0.8448	0.8769	+0.0321

The P2 baseline shows consistent improvement across all metrics. While the overall mAP@0.5 improvement appears modest (+0.9%), the weighted mAP@0.5 reveals a more substantial +3.2% gain, indicating that the improvements are concentrated on the most numerous object categories.

Metrics by Object Size



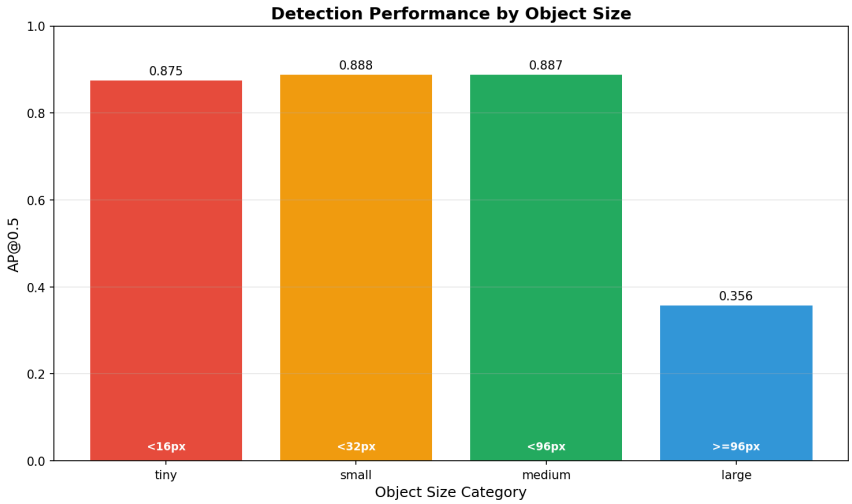


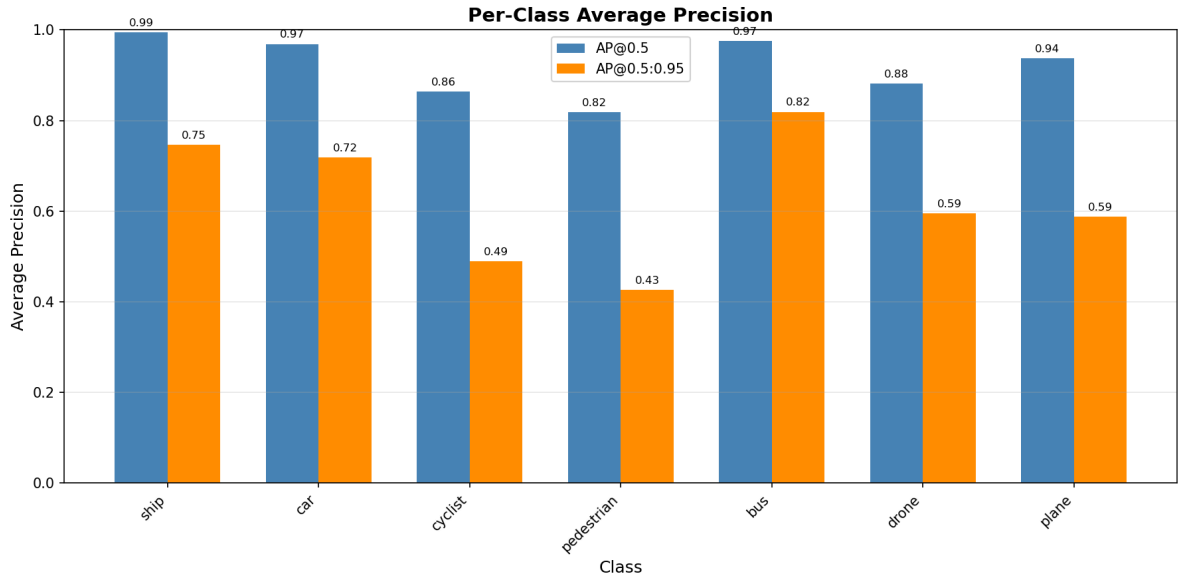
Figure Naive vs P2-mod object size

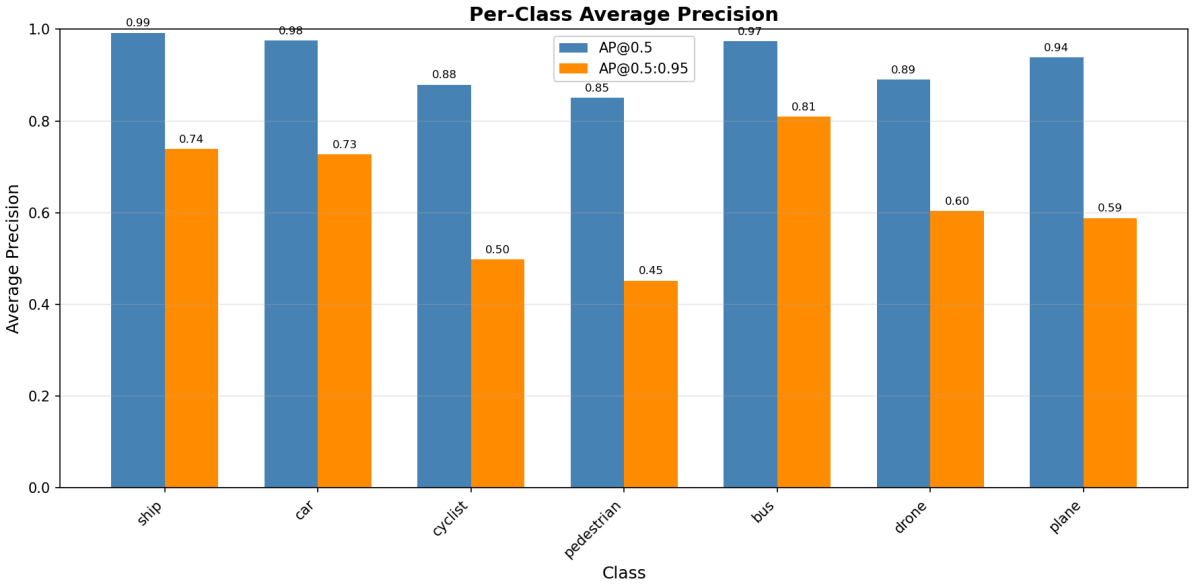
Metrics by Size (AP@0.5)			
Size Category	**Naive**	**P2-mod**	**Improvement**
Tiny (< 16px)	0.8335	0.8747	+0.0412
Small (< 32px)	0.8920	0.8881	-0.0039
Medium (< 96px)	0.8904	0.8875	-0.0029
Large (>= 96px)	0.9505	0.3564	-0.5941

The P2 head delivers a significant +4.1% AP improvement on tiny objects, which represent 80.6% of the dataset. The small and medium categories show negligible changes (-0.4% and -0.3%), which is expected since the P3 head already handles these sizes adequately. The large drop in the large category (-59.4%) is not statistically significant, as it is based on only 48 ground truth objects in the validation set.

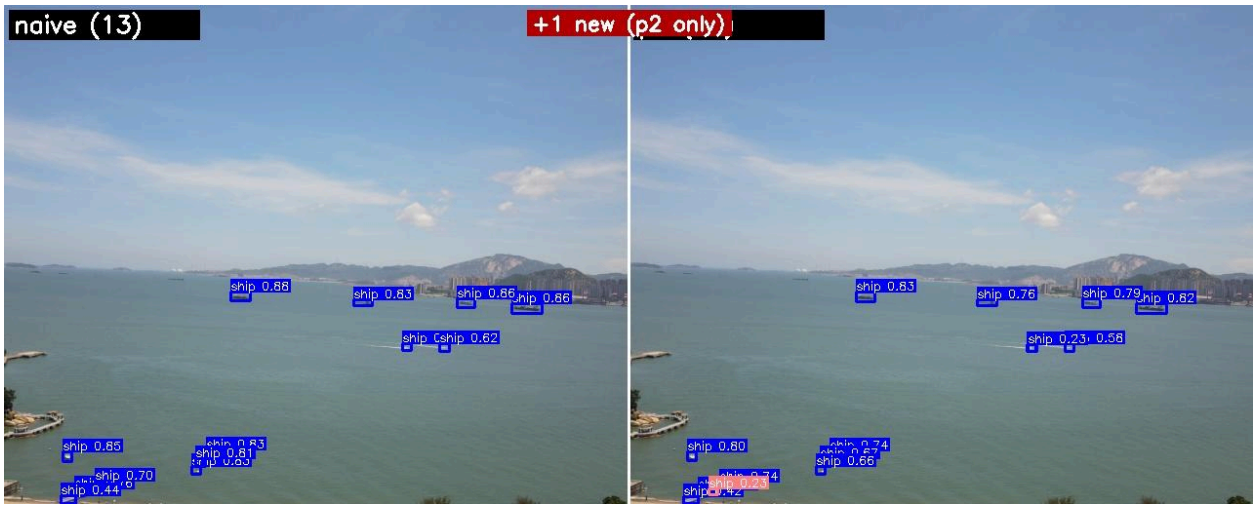
Per Class AP@0.5			
Object Class	**Naive Baseline Performance**	**P2 Baseline Performance**	**Performance Improvement**
Ship	**0.9944**	0.9924	-0.0020
Car	0.9688	**0.9759**	+0.0071
Cyclist	0.8633	**0.8793**	+0.0160
Pedestrian	0.8187	**0.8504**	+0.0317
Bus	**0.9750**	0.9745	-0.0005
Drone	0.8820	**0.8906**	+0.0086
Plane	0.9368	**0.9392**	+0.0024

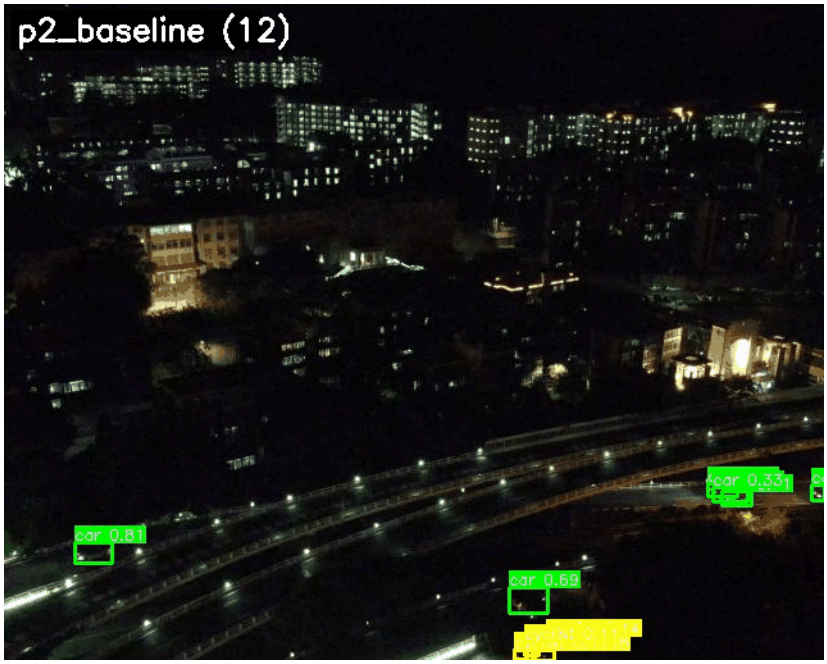
The P2 head benefits pedestrians (+3.2%) and cyclists (+1.6%) the most. These are typically the smallest object classes in the dataset, confirming that the stride-4 detection head specifically improves detection of the most challenging tiny targets.





Quality Metrics from P2-mod





Experiment SAHI inference on top of P2-mod

The **stage 1** and **stage 1b** grid search can be summed into one key discovery that At conf=0.25 with slice=512, SAHI outperformed standard inference for the first time. So we will present the results from **stage 2** experiments.

Rank	Configuration	mAP@0.5	Weighted mAP@0.5	Tiny AP	Small AP	vs Std W-mAP
1	GREEDYNMM, IOU, t=0.7	0.8946	0.8630	0.8513	0.9136	+0.0290
2	NMS, IOS, t=0.7	0.8947	0.8622	0.8554	0.8913	+0.0282
3	NMS, IOU, t=0.5	0.9006	0.8614	0.8527	0.8992	+0.0274
4	NMS, IOU, t=0.7	0.8945	0.8606	0.8520	0.8982	+0.0266

5	GREEDYNMM, IOS, t=0.7	0.8938	0.8570	0.8436	0.9147	+0.0229
6	NMS, IOS, t=0.5	0.8946	0.8554	0.8469	0.8912	+0.0214
7	GREEDYNMM, IOU, t=0.5	0.8912	0.8552	0.8417	0.9135	+0.0212
8	NMS, IOU, t=0.3	0.8942	0.8549	0.8464	0.8911	+0.0209
9	GREEDYNMM, IOU, t=0.3	0.8833	0.8481	0.8329	0.9133	+0.0141
10	GREEDYNMM, IOS, t=0.5	0.8836	0.8469	0.8333	0.9044	+0.0129
11	NMS, IOS, t=0.3	0.8657	0.8333	0.8275	0.8518	-0.0007
12	GREEDYNMM, IOS, t=0.3	0.8360	0.7988	0.7928	0.8252	-0.0352

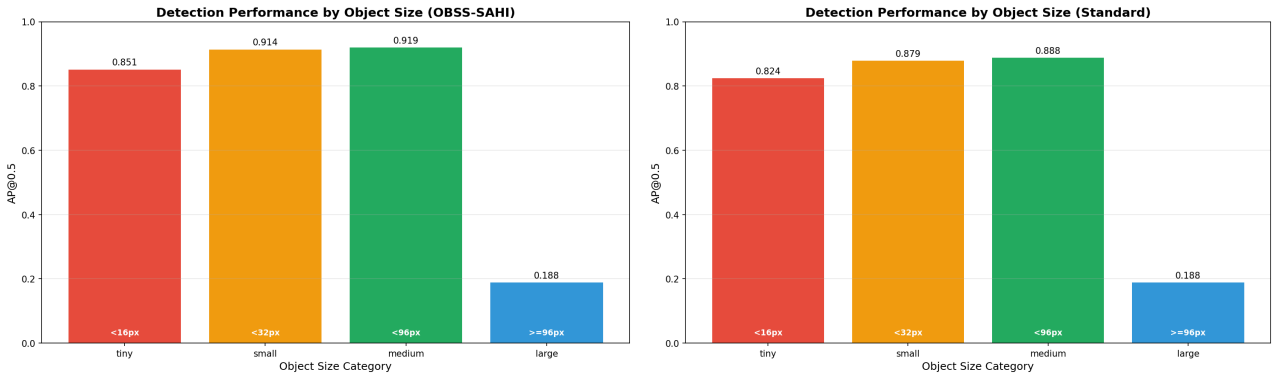
Winner: GREEDYNMM with IOU metric at threshold 0.7, achieving the highest weighted mAP@0.5 of 0.8630 (+2.9% over standard).

Results below compare standard P2 inference vs P2 + SAHI inference at the same confidence threshold (0.25).

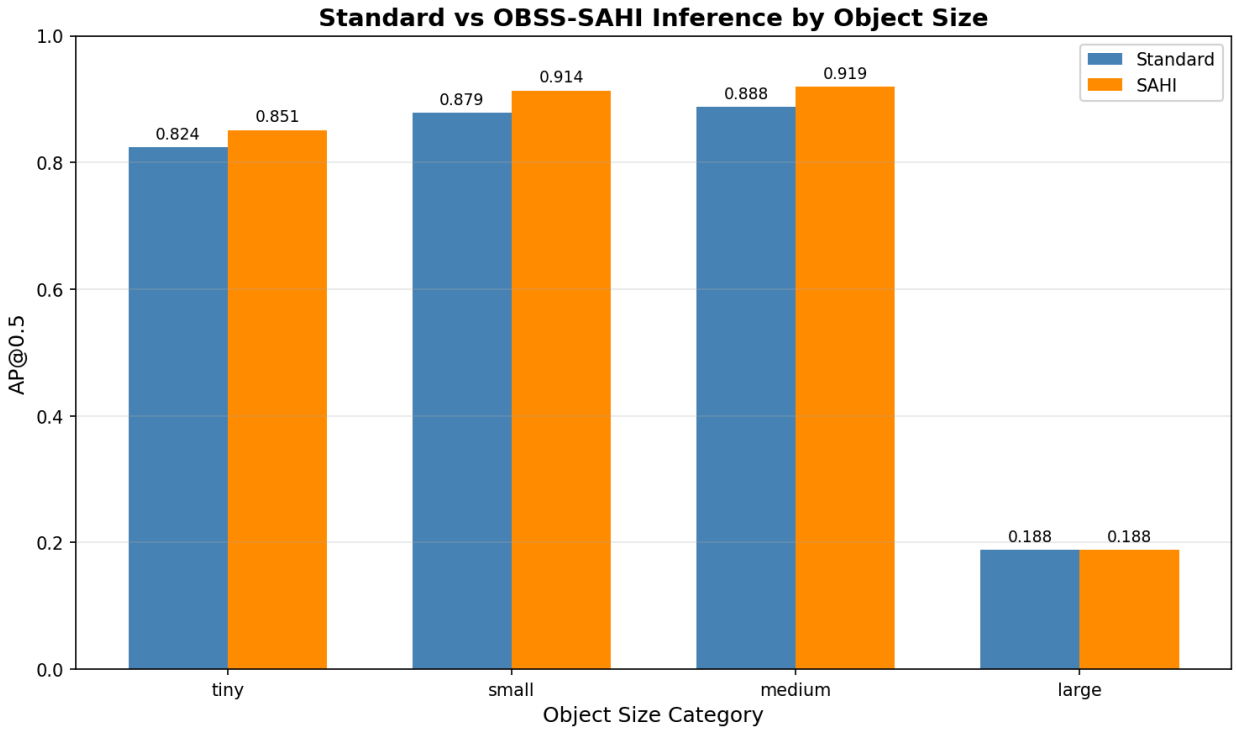
Size breakdown — Standard vs SAHI (P2 model, conf=0.25)			
Size Category	**Standard Metric**	**SAHI Metric**	**Improvement Achieved**
Tiny (less than 16px)	0.8236	**0.8513**	**+0.0277**
Small (less than 32px)	0.8789	**0.9136**	**+0.0347**
Medium (less than 96px)	0.8876	**0.9193**	**+0.0317**
Large (greater than or equal to 96px)	0.1881	0.1881	0.0000
Weighted mAP@0.5	**0.8340**	**0.8630**	**+0.0290**

SAHI improves detection across all size categories except large (unchanged). The strongest gains are on **small** (+3.5%) and **medium** (+3.2%) objects, with **tiny** objects also seeing a meaningful +2.8% improvement. The total prediction count increases moderately from 63,250 to 75,116 — a controlled 19% increase compared to the catastrophic explosion seen at lower confidence thresholds.

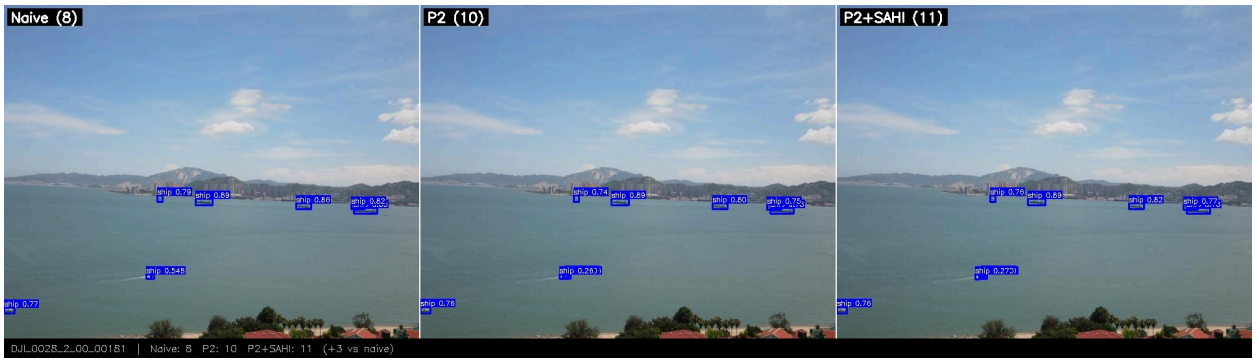
Overall Metric SAHI vs P2-mod



Metrics by Size



Quality Metrics



FPS benchmark per model

Model	**FPS**
Naive	53.1 +-6
P2-mod	48.1 +-3.8

P2-mod plus Sahi	9.5 +- 0.5

We notice that SAHI comes at a significant inference speed cost. FPS benchmarks reveal the trade-off between accuracy and speed.
The P2 head adds a modest ~20% overhead while delivering the largest accuracy gain. SAHI, on the other hand, reduces throughput by ~80% (from 48.1 to 9.5 FPS), dropping well below the 30 FPS real-time threshold. This makes SAHI unsuitable for real-time applications but viable for offline batch processing where accuracy is prioritized over speed.

Key Insights

1. **P2 head and SAHI are complementary but address the same problem differently.** The P2 head improves feature resolution during training (stride 4 vs stride 8), while SAHI improves input resolution during inference (patches vs full image). The gains do not fully stack — the P2 head already captures much of what SAHI would recover — but the combination still produces the best overall results.
2. **SAHI's effectiveness depends on the input resolution**. The original SAHI paper reports 5-10% AP improvements on datasets like VisDrone and xView, which contain images at 2000x1500 pixels or higher. These images undergo significant downscaling when fed to a 640-pixel detector, losing critical detail for small objects. Our 640x512 images are already at the detector's native resolution, explaining the comparatively smaller SAHI gains. On higher-resolution aerial imagery, the benefit of SAHI would be substantially larger.

References

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3. Wang, J., Xu, C., Yang, W., & Yu, L. (2022). A Normalized Gaussian Wasserstein Distance for Tiny Object Detection.

   https://arxiv.org/abs/2110.13389

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4. SOD-YOLOv8: Li, Y. et al. (2024). SOD-YOLOv8 —. https://arxiv.org/html/2408.04786v1